Block practice · Mobile networks, IP addresses and internet hardware

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Mobile generations: 3G, 4G and 5G

Connecting to the internet: switch, WAP, router and modem

IPv4 and IPv6: sizes, formats and why IPv6 exists

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Practice workbook · Mobile generations: 3G, 4G and 5G

Characteristics and benefits of newer mobile generations, including higher frequency bands, bandwidth and latency.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-06

1. Question 3 · part d.ii · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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1. Question 3 · part d.ii · 2 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d.ii only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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1. Question 3 · part d.ii · 2 marks · Working and answer

Working and answer

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2. Question 3 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

	Layer name	Function
1		
2		

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2. Question 3 · part d.i · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part d.i only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

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- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
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(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

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(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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2. Question 3 · part d.i · 1 mark · Working and answer

Working and answer

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3. Question 1 · part c · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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3. Question 1 · part c · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c only.

(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

.....

2 .....

.....

(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

.....

.....

(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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3. Question 1 · part c · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part c only.

(d) Programmers write and test code.

They use modular testing.

State what is meant by the term **modular testing**.

(1)

(e) Computer scientists are viewed as professionals, in the same way that doctors are viewed as professionals.

Give **two** ways that computer scientists can demonstrate professionalism.

(2)

1 .....

2 .....

(Total for Question 1 = 10 marks)

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Working and answer

4. Question 4 · part a.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a.i only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

.....

.....

.....

.....

(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

.....

2 .....

.....

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4. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Connecting to the internet: switch, WAP, router and modem

Placing and linking internet-access components in a diagram, wired or wireless, in the correct order towards the ISP/internet.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-13

1. Question 1 · part d · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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2

P 7 8 6 3 7 A 0 2 2 0

1. Question 1 · part d · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

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(d) A library sets up a network using a star topology.

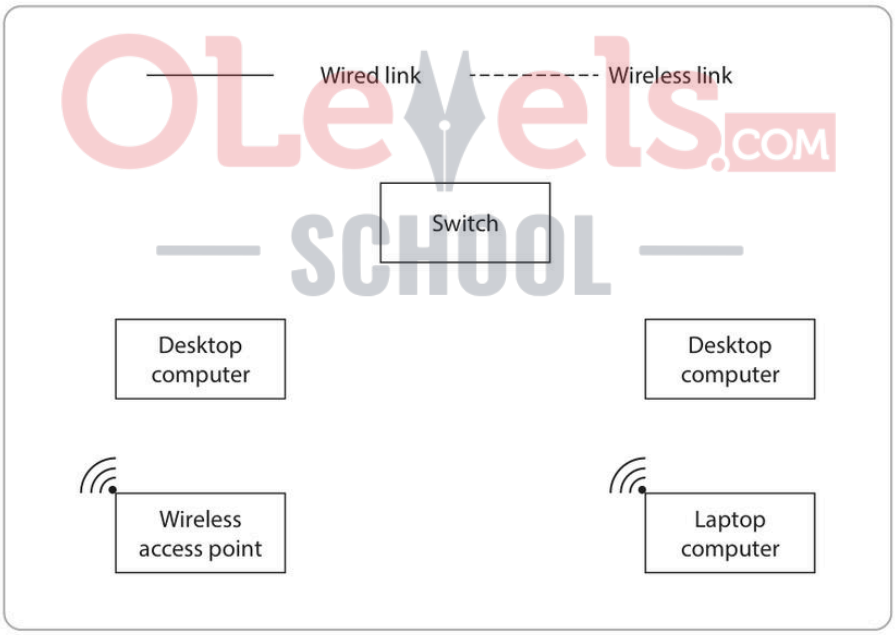
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

(3)



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1. Question 1 · part d · 3 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part d only.

(e) IT teams carry out network penetration testing.

Describe the purpose of penetration testing.

(2)

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(Total for Question 1 = 12 marks)

Working and answer

OLevels.COM SCHOOL

2. Question 2 · part d · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

2 Zafer and Robert work for a company that makes washing machines.

(a) Zafer writes user manuals for the washing machines.

He stores these documents in the cloud.

Zafer and the cloud storage provider share responsibility for data security.

State **one** area of responsibility for each of them.

(2)

Zafer

.....

.....

Cloud storage provider

.....

.....

(b) Robert programs the robots that make the parts for the washing machines.

Zafer writes the user manuals.

Robert and Zafer have different levels of access to folders and files on the company network.

The different levels of access are Read, Write, Execute and None.

Complete the table to show the levels of access that each has.

(2)

	Robert	**Zafer**
Washing machine design drawings		
A folder of manuals for new machines		

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P 6 1 8 8 0 A 0 4 2 4

2. Question 2 · part d · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

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(c) Zafer uses a browser to view pages on Pearson's website by typing a uniform resource locator into a browser. This is shown in **Figure 1**.



Figure 1

Pearson's machine	Zafer's machine
MAC address: 03-57-00-EC-4B-30	MAC address: 00-15-00-BC-9A-90
IPv4 address: 2.20.38.113	IPv4 address: 192.168.1.78
IPv4 Subnet Mask: 255.255.0.0	IPv4 Subnet Mask: 255.255.255.0
Website domain: www.pearson.com	Web browser
Main web page: https://www.pearson.com/uk/	Uniform resource locator: www.pearson.com

A domain name server is used in this process.

Identify the input to and the output from the domain name server.

(2)

Input

Output

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2. Question 2 · part d · 6 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

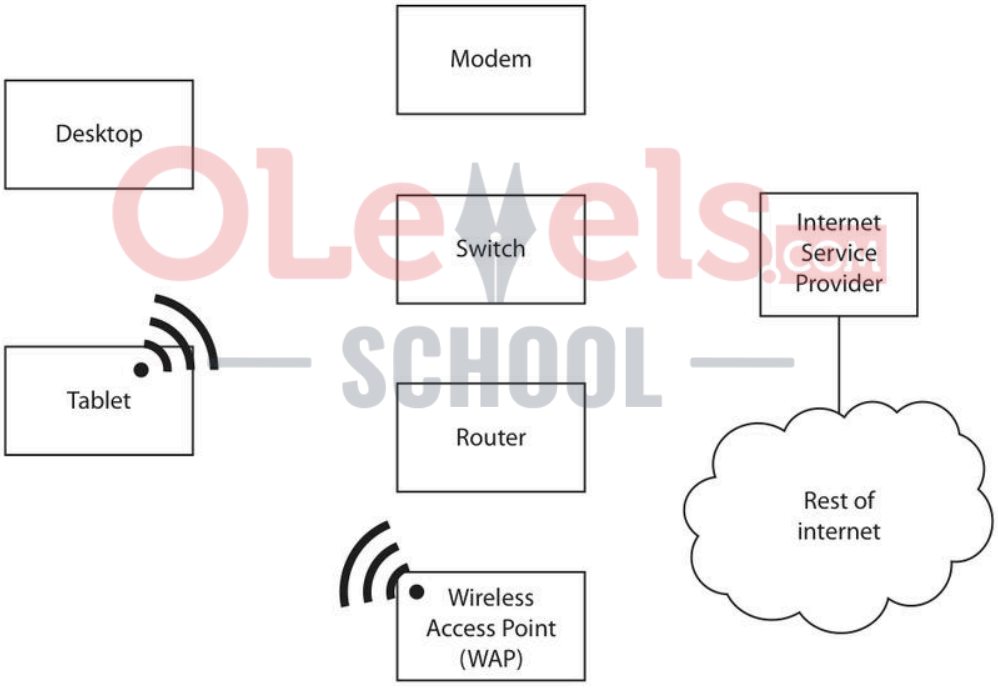
(d) Zafer can use a desktop or a tablet computer to connect to the internet.

Only the tablet has wireless capabilities.

The diagram shows the components used to connect to the internet.

Complete the diagram to show how the desktop and the tablet are connected to the internet.

(6)



(Total for Question 2 = 12 marks)

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2. Question 2 · part d · 6 marks · Working and answer

Working and answer

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3. Question 4 · part d · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

OLevels.com SCHOOL

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

3. Question 4 · part d · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

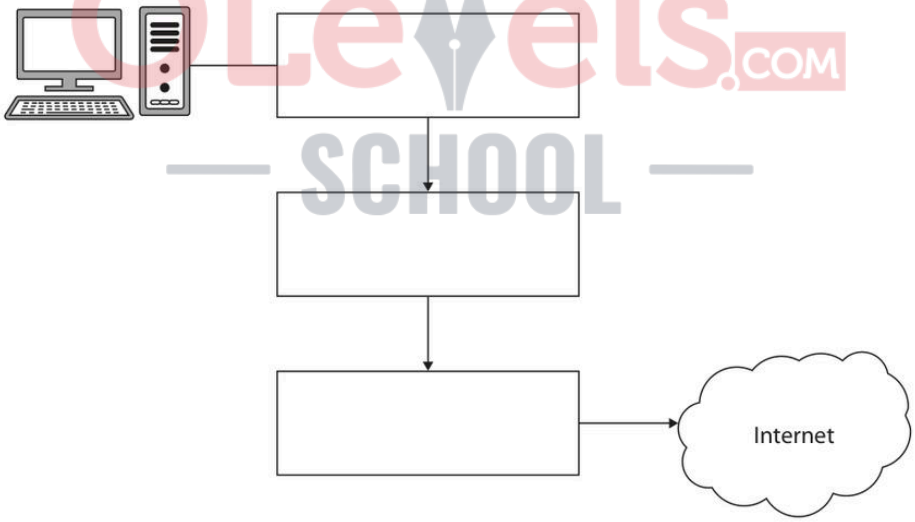
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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12

P 7 2 9 3 7 A 0 1 2 2 4

3. Question 4 · part d · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

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(e) Users enter passwords when logging onto a network and when creating accounts using a web page.

Describe **one** difference between validation and authentication.

(2)

(Total for Question 4 = 12 marks)

Working and answer

OLevels.com SCHOOL

Practice workbook · IPv4 and IPv6: sizes, formats and why IPv6 exists

Why IPv6 was introduced, bit lengths of IPv4 and IPv6, and relating the hexadecimal group structure of an IPv6 address to its 128-bit size.

3 question parts: 1 worked-example retry and 2 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-12

1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

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(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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1. Question 4 · part e · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part e only.

(f) The research facility is considering introducing cameras and facial recognition software to monitor the work and movement of people on and around its site.

Discuss the privacy issues that need to be considered when deciding whether or not to install this new system.

(6)

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(Total for Question 4 = 16 marks)

1. Question 4 · part e · 3 marks · Working and answer

Working and answer

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2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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2. Question 1 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

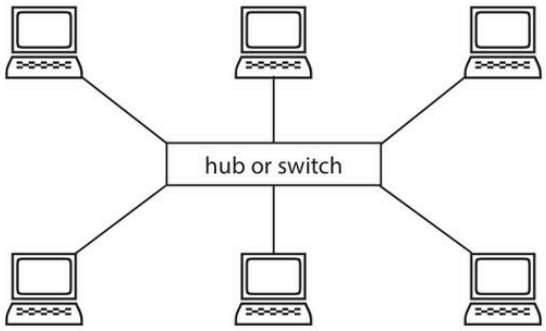


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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P 6 1 8 8 4 R R A 0 4 2 0

2. Question 1 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part b · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

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3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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